OUTPUT SCREENSHOTS

QUERY -1 :

What are the pricing plans available for your software?

```
(venv) PS D:\Downloads\customer_support_system\customer_support_system> python demo.py
[Memory] SQLite database ready at: D:\Downloads\customer_support_system\customer_support_system\memory.db

[System] Starting AI Customer Support System...
[Graph] LangGraph workflow compiled successfully.
[System] System ready.


================================================================

  ABC TECHNOLOGIES — AI CUSTOMER SUPPORT SYSTEM
  Built with LangGraph

================================================================

  Please enter your details to begin.

  Your name        : Harshita
  Your customer ID: 1

  Welcome, Harshita! (ID: 1)
  Type your support query below. Type 'quit' to exit.

----------------------------------------------------------------
  Your query : What are the pricing plans available for your software

[System] Processing your query...


[Intent] Classifying query: 'What are the pricing plans available for your software'
[Memory] Retrieved 0 history entries for '1'.
[Intent] Classified as: Sales
[Router] Intent='Sales' → routing to 'sales_agent'
[Sales Agent] Processing query...
[RAG] Loaded 4 knowledge-base documents.
[RAG] Created 30 text chunks.
[RAG] Loading HuggingFace embedding model (all-MiniLM-L6-v2)...
Warning: You are sending unauthenticated requests to the HF Hub. Please set a HF_TOKEN to enable higher rate limits and faster down
loads.
Loading weights: 100%|▰▰▰▰▰▰▰▰▰▰▰▰▰▰▰▰▰▰▰▰▰▰▰▰▰▰▰▰▰▰▰▰▰▰▰▰▰▰▰▰▰▰▰▰▰▰▰▰▰▰▰▰▰▰| 103/103 [00:00<00:00, 3341.58it/s]
[RAG] FAISS vector store built successfully.
[RAG] Retrieved 4 context chunks for query.

[Supervisor] Reviewing draft response...
```

```
PROBLEMS    DEBUG CONSOLE    PORTS    TERMINAL                                   python

[Supervisor] Reviewing draft response...
[Supervisor] Draft passes all quality checks ✓
[Supervisor] Notes: Draft approved without modifications.
[Memory] Saved interaction for customer '1'.
[Supervisor] Interaction saved to memory for customer '1'.

================================================================

  Customer   : Harshita (1)
  Query      : What are the pricing plans available for your software
  Intent     : Sales
  Department : Sales
  Supervisor : Draft approved without modifications.
----------------------------------------------------------------
  RESPONSE:
----------------------------------------------------------------
  Hello Harshita,

  Thank you for contacting ABC Technologies Sales.

  I understand your query: "What are the pricing plans available for your software"

  Based on our knowledge base, here is the information relevant to your request:

  [Source: pricing_guide]
  ABC TECHNOLOGIES - PRICING GUIDE
  ================================

  1. SUBSCRIPTION PLANS
  ---------------------

  STARTER PLAN - $29/month (or $290/year - save 17%)
  - Up to 5 users
  - 10 GB cloud storage
  - Basic reporting
  - Email support
  - Core business management features
  - Mobile app access

  ---

  [Source: pricing_guide]
  PROFESSIONAL PLAN - $79/month (or $790/year - save 17%)
  - Up to 25 users
```

PROBLEMS DEBUG CONSOLE PORTS TERMINAL python

```
[Source: pricing_guide]
PROFESSIONAL PLAN - $79/month (or $790/year - save 17%)
- Up to 25 users
- 50 GB cloud storage
- Advanced reporting & analytics
- Priority email & chat support
- All Starter features
- API access
- Custom integrations
- Automated workflows

---

[Source: pricing_guide]
ENTERPRISE PLAN - $199/month (or $1,990/year - save 17%)
- Unlimited users
- 500 GB cloud storage
- Custom reporting
- 24/7 dedicated support
- All Professional features
- Custom SLA
- On-premise deployment option
- Dedicated account manager
- SSO / SAML integration
- Advanced security features

---

[Source: pricing_guide]
2. ADD-ONS
-----------
- Extra Storage: $10/month per 10 GB
- Additional Users (Starter): $5/user/month
- Advanced Analytics Module: $30/month
- Custom Branding: $20/month
- Priority Support Upgrade: $25/month

3. FREE TRIAL
--------------
- 14-day free tri


Please do not hesitate to reach out if you need further assistance.
```

```
Please do not hesitate to reach out if you need further assistance.

Best regards,
Alex | Sales Specialist
ABC Technologies Sales

================================================================

Do you have another query? (yes/no) : no

================================================================

Thank you for contacting ABC Technologies, Harshita!
Total queries handled this session: 1
Your conversation history is saved under ID: 1

================================================================

(venv) PS D:\Downloads\customer_support_system\customer_support_system>
```

QUERY -2 :

I forgot my account password.

```
PROBLEMS    DEBUG CONSOLE    PORTS    TERMINAL                                   powershell

(venv) PS D:\Downloads\customer_support_system\customer_support_system> python demo.py
[Memory] SQLite database ready at: D:\Downloads\customer_support_system\customer_support_system\memory.db

[System] Starting AI Customer Support System...
[Graph] LangGraph workflow compiled successfully.
[System] System ready.


================================================================

  ABC TECHNOLOGIES — AI CUSTOMER SUPPORT SYSTEM
  Built with LangGraph

================================================================

  Please enter your details to begin.

  Your name       : Harshita
  Your customer ID: 1

  Welcome, Harshita! (ID: 1)
  Type your support query below. Type 'quit' to exit.

----------------------------------------------------------------
  Your query : I forgot my password

[System] Processing your query...


[Intent] Classifying query: 'I forgot my password'
[Memory] Retrieved 6 history entries for '1'.
[Intent] Classified as: Account
[Router] Intent='Account' → routing to 'account_agent'
[Account Agent] Processing query...
[RAG] Loaded 'company_policy' as 1 whole-file document.
[RAG] Loaded 'pricing_guide' as 1 whole-file document.
[RAG] Loaded 'technical_manual' as 1 whole-file document.
[RAG] Loaded 'faq' as 28 Q&A chunks.
[RAG] Loaded 31 total knowledge-base documents/chunks.
[RAG] 28 pre-chunked (FAQ) docs + 22 character-split chunks = 50 total chunks.
```

```
PROBLEMS    DEBUG CONSOLE    PORTS    TERMINAL                                   powershell

Warning: You are sending unauthenticated requests to the HF Hub. Please set a HF_TOKEN to enable higher rate limits and faster down
loads.
Loading weights: 100%|▰▰▰▰▰▰▰▰▰▰▰▰▰▰▰▰▰▰▰▰▰▰▰▰▰▰▰▰▰▰▰▰▰▰▰▰▰▰▰▰▰▰▰▰▰▰▰▰▰▰▰▰▰▰| 103/103 [00:00<00:00, 4024.94it/s]
[RAG] FAISS vector store built successfully.
[RAG] Retrieved 1/4 chunks above threshold (0.5) for query: 'I forgot my password'
       [kept] score=0.6140 source=faq
       [dropped] score=0.9390 source=faq
       [dropped] score=1.0329 source=technical_manual
       [dropped] score=1.4150 source=faq

[Supervisor] Reviewing draft response...
[Supervisor] Draft passes all quality checks ✓
[Supervisor] Notes: Draft approved without modifications.
[Memory] Saved interaction for customer '1'.
[Supervisor] Interaction saved to memory for customer '1'.

================================================================

  Customer   : Harshita (1)
  Query      : I forgot my password
  Intent     : Account
  Department : Account Management
  Supervisor : Draft approved without modifications.
----------------------------------------------------------------
  RESPONSE:
----------------------------------------------------------------
  Hello Harshita,

  Thank you for contacting ABC Technologies Account Management.

  I understand your query: "I forgot my password"

  Based on our knowledge base, here is the information relevant to your request:

  [Source: faq]
  Q: How do I reset my password?
  A: Click "Forgot Password" on the login page. A reset link will be emailed to your registered address. The link expires in 24 hou
rs.

  Please do not hesitate to reach out if you need further assistance.
```

```
    Best regards,
    Riley | Account Support Specialist
    ABC Technologies Account Management

  ================================================================

    Do you have another query? (yes/no) : no

  ================================================================

    Thank you for contacting ABC Technologies, Harshita!
    Total queries handled this session: 1
    Your conversation history is saved under ID: 1

  ================================================================

  (venv) PS D:\Downloads\customer_support_system\customer_support_system>
```

QUERY -3 :

My application crashes whenever I upload a file.

```
(venv) PS D:\Downloads\customer_support_system\customer_support_system> python demo.py
[Memory] SQLite database ready at: D:\Downloads\customer_support_system\customer_support_system\memory.db

[System] Starting AI Customer Support System...
[Graph] LangGraph workflow compiled successfully.
[System] System ready.


================================================================

  ABC TECHNOLOGIES — AI CUSTOMER SUPPORT SYSTEM
  Built with LangGraph

================================================================

  Please enter your details to begin.

  Your name       : Raju
  Your customer ID: 2

  Welcome, Raju! (ID: 2)
  Type your support query below. Type 'quit' to exit.

----------------------------------------------------------------
  Your query : My application crashes whenever I upload a file.

[System] Processing your query...


[Intent] Classifying query: 'My application crashes whenever I upload a file.'
[Memory] Retrieved 0 history entries for '2'.
[Intent] Classified as: Technical
[Router] Intent='Technical' → routing to 'technical_agent'
[Technical Agent] Processing query...
[RAG] Loaded 'company_policy' as 1 whole-file document.
[RAG] Loaded 'pricing_guide' as 1 whole-file document.
[RAG] Loaded 'technical_manual' as 1 whole-file document.
[RAG] Loaded 'faq' as 28 Q&A chunks.
[RAG] Loaded 31 total knowledge-base documents/chunks.
[RAG] 28 pre-chunked (FAQ) docs + 22 character-split chunks = 50 total chunks.
[RAG] Loading HuggingFace embedding model (all-MiniLM-L6-v2)...
Warning: You are sending unauthenticated requests to the HF Hub. Please set a HF_TOKEN to enable higher rate limits and faster down
loads.
Loading weights: 100%|██████████████████████████████████████████████████████| 103/103 [00:00<00:00, 7389.90it/s]
[RAG] FAISS vector store built successfully.
[RAG] Retrieved 1/4 chunks above threshold (0.5) for query: 'My application crashes whenever I upload a file.'
       [kept] score=0.5095 source=faq
       [dropped] score=0.6924 source=technical_manual
```

```
        [dropped] score=0.8772 source=technical_manual
        [dropped] score=1.1376 source=faq
[RAG] Retrieved 1/2 chunks above threshold (0.5) for query: 'file upload troubleshooting application error'
        [kept] score=0.6897 source=technical_manual
        [dropped] score=0.8142 source=technical_manual

[Supervisor] Reviewing draft response...
[Supervisor] Draft passes all quality checks ✓
[Supervisor] Notes: Draft approved without modifications.
[Memory] Saved interaction for customer '2'.
[Supervisor] Interaction saved to memory for customer '2'.

==================================================================

  Customer   : Raju (2)
  Query      : My application crashes whenever I upload a file.
  Intent     : Technical
  Department : Technical Support
  Supervisor : Draft approved without modifications.
------------------------------------------------------------------
  RESPONSE:
------------------------------------------------------------------
  Hello Raju,

  Thank you for contacting ABC Technologies Technical Support.

  I understand your query: "My application crashes whenever I upload a file."

  Based on our knowledge base, here is the information relevant to your request:

  [Source: faq]
  Q: Why is my application crashing?
  A: Common causes include browser cache issues, unsupported file formats, or files exceeding size limits. Try clearing your cache
or using a different browser.

  --- Additional Troubleshooting ---
  [Source: technical_manual]
  4. FILE UPLOAD TROUBLESHOOTING
  -------------------------------
  Common causes of file upload crashes:
  - File size exceeds the plan limit (Starter: 25MB, Pro: 100MB, Enterprise: 500MB)
  - Unsupported file format (supported: PDF, DOCX, XLSX, PNG, JPG, CSV, ZIP)
  - Slow internet connection causing timeout
  - Browser cache/cookies issue

  Please do not hesitate to reach out if you need further assistance.

  Best regards,
```

```
  Best regards,
  Sam | Technical Support Engineer
  ABC Technologies Technical Support

==================================================================

  Do you have another query? (yes/no) : no

==================================================================

  Thank you for contacting ABC Technologies, Raju!
  Total queries handled this session: 1
  Your conversation history is saved under ID: 2

==================================================================

(venv) PS D:\Downloads\customer_support_system\customer_support_system>
```

QUERY -4:

I need a refund for my annual subscription.

```
(venv) PS D:\Downloads\customer_support_system\customer_support_system> python demo.py
[Memory] SQLite database ready at: D:\Downloads\customer_support_system\customer_support_system\memory.db

[System] Starting AI Customer Support System...
[Graph] LangGraph workflow compiled successfully.
[System] System ready.


================================================================

  ABC TECHNOLOGIES — AI CUSTOMER SUPPORT SYSTEM
  Built with LangGraph

================================================================

  Please enter your details to begin.

  Your name       : Pranay
  Your customer ID: 3

  Welcome, Pranay! (ID: 3)
  Type your support query below. Type 'quit' to exit.

----------------------------------------------------------------
  Your query :  I need a refund for my annual subscription.

[System] Processing your query...


[Intent] Classifying query: 'I need a refund for my annual subscription.'
[Memory] Retrieved 0 history entries for '3'.
[Intent] Classified as: Billing
[Router] Intent='Billing' → routing to 'billing_agent'
[Billing Agent] Processing query...
[RAG] Loaded 'company_policy' as 1 whole-file document.
[RAG] Loaded 'pricing_guide' as 1 whole-file document.
[RAG] Loaded 'technical_manual' as 1 whole-file document.
[RAG] Loaded 'faq' as 28 Q&A chunks.
[RAG] Loaded 31 total knowledge-base documents/chunks.
[RAG] 28 pre-chunked (FAQ) docs + 22 character-split chunks = 50 total chunks.
[RAG] Loading HuggingFace embedding model (all-MiniLM-L6-v2)...
Warning: You are sending unauthenticated requests to the HF Hub. Please set a HF_TOKEN to enable higher rate limits and faster down
loads.
Loading weights: 100%|                                                          | 103/103 [00:00<00:00, 7761.93it/s]
[RAG] FAISS vector store built successfully.
[RAG] Retrieved 1/4 chunks above threshold (0.5) for query: 'I need a refund for my annual subscription.'
       [kept] score=0.9599 source=faq
        [dropped] score=1.1980 source=faq
 [Billing Agent] ⚠️ HIGH-RISK request detected — flagging for human approval.
 [Router] High-risk request detected → human approval required.

 ================================================================
 🔴  HUMAN-IN-THE-LOOP APPROVAL REQUIRED
 ================================================================
 Customer ID  : 3
 Customer Name: Pranay
 Intent       : Billing
 Department   : Billing

 Customer Query:
   I need a refund for my annual subscription.

 Draft Response (for supervisor review):
 ----------------------------------------------------------------
 Hello Pranay,

 Thank you for contacting ABC Technologies Billing Support.

 I have received your request regarding: "I need a refund for my annual subscription."

 This request requires review and approval from our billing supervisor before it can be processed. Your case has been escalated and
 you will receive a response within 24 business hours.

 Reference information from our policy:

 [Source: faq]
 Q: How do I request a refund?
 A: Submit a refund request through the support portal. Refunds within 30 days are processed automatically; others are reviewed manu
 ally within 5 business days.

 Best regards,
 Jordan | Billing Support Specialist
 ABC Technologies Billing
 ----------------------------------------------------------------

 Supervisor Decision — type 'approve' or 'reject': approve

 [Human Approval] Decision recorded: APPROVED
 ================================================================

 [Router] Request approved by supervisor → continuing to response.

 [Supervisor] Reviewing draft response...
 [Supervisor] Draft passes all quality checks ✓
 [Supervisor] Notes: Draft approved without modifications.
```

```
[Memory] Saved interaction for customer '3'.
[Supervisor] Interaction saved to memory for customer '3'.

================================================================

  Customer   : Pranay (3)
  Query      : I need a refund for my annual subscription.
  Intent     : Billing
  Department : Billing
  Approval   : APPROVED
  Supervisor : Draft approved without modifications.
----------------------------------------------------------------
  RESPONSE:
----------------------------------------------------------------
  Hello Pranay,

  Thank you for contacting ABC Technologies Billing Support.

  I have received your request regarding: "I need a refund for my annual subscription."

  This request requires review and approval from our billing supervisor before it can be processed. Your case has been escalated an
d you will receive a response within 24 business hours.

  Reference information from our policy:

  [Source: faq]
  Q: How do I request a refund?
  A: Submit a refund request through the support portal. Refunds within 30 days are processed automatically; others are reviewed ma
nually within 5 business days.

  Best regards,
  Jordan | Billing Support Specialist
  ABC Technologies Billing

================================================================

  Do you have another query? (yes/no) : no

================================================================

  Thank you for contacting ABC Technologies, Pranay!
  Total queries handled this session: 1
  Your conversation history is saved under ID: 3

================================================================

(venv) PS D:\Downloads\customer_support_system\customer_support_system>
```

Human Approval-rejected

```
(venv) PS D:\Downloads\customer_support_system\customer_support_system> python demo.py
[Memory] SQLite database ready at: D:\Downloads\customer_support_system\customer_support_system\memory.db

[System] Starting AI Customer Support System...
[Graph] LangGraph workflow compiled successfully.
[System] System ready.


================================================================

  ABC TECHNOLOGIES — AI CUSTOMER SUPPORT SYSTEM
  Built with LangGraph

================================================================

  Please enter your details to begin.

  Your name       : Krishna
  Your customer ID: 4

  Welcome, Krishna! (ID: 4)
  Type your support query below. Type 'quit' to exit.

----------------------------------------------------------------
  Your query :    I need a refund for my annual subscription.

[System] Processing your query...


[Intent] Classifying query: 'I need a refund for my annual subscription.'
[Memory] Retrieved 0 history entries for '4'.
[Intent] Classified as: Billing
[Router] Intent='Billing' → routing to 'billing_agent'
[Billing Agent] Processing query...
[RAG] Loaded 'company_policy' as 1 whole-file document.
[RAG] Loaded 'pricing_guide' as 1 whole-file document.
[RAG] Loaded 'technical_manual' as 1 whole-file document.
[RAG] Loaded 'faq' as 28 Q&A chunks.
[RAG] Loaded 31 total knowledge-base documents/chunks.
[RAG] 28 pre-chunked (FAQ) docs + 22 character-split chunks = 50 total chunks.
[RAG] Loading HuggingFace embedding model (all-MiniLM-L6-v2)...
Warning: You are sending unauthenticated requests to the HF Hub. Please set a HF_TOKEN to enable higher rate limits and faster down
loads.
Loading weights: 100%|                                                    | 103/103 [00:00<00:00, 7029.52it/s]
[RAG] FAISS vector store built successfully.
[RAG] Retrieved 1/4 chunks above threshold (0.5) for query: 'I need a refund for my annual subscription.'
       [kept] score=0.9599 source=faq
```

```
================================================================
🔴 HUMAN-IN-THE-LOOP APPROVAL REQUIRED
================================================================
Customer ID  : 4
Customer Name: Krishna
Intent       : Billing
Department   : Billing

Customer Query:
  I need a refund for my annual subscription.

Draft Response (for supervisor review):
----------------------------------------------------------------
Hello Krishna,

Thank you for contacting ABC Technologies Billing Support.

I have received your request regarding: "I need a refund for my annual subscription."

This request requires review and approval from our billing supervisor before it can be processed. Your case has been escalated and
you will receive a response within 24 business hours.

Reference information from our policy:

[Source: faq]
Q: How do I request a refund?
A: Submit a refund request through the support portal. Refunds within 30 days are processed automatically; others are reviewed manu
ally within 5 business days.

Best regards,
Jordan | Billing Support Specialist
ABC Technologies Billing
----------------------------------------------------------------

Supervisor Decision — type 'approve' or 'reject': reject

[Human Approval] Decision recorded: REJECTED
================================================================

[Router] Request rejected by supervisor → ending workflow.
[End] Request was rejected by supervisor. Sending rejection response.
[Memory] Saved interaction for customer '4'.

================================================================

  Customer   : Krishna (4)
  Query      : I need a refund for my annual subscription.
```

```
[Human Approval] Decision recorded: REJECTED
================================================================

[Router] Request rejected by supervisor → ending workflow.
[End] Request was rejected by supervisor. Sending rejection response.
[Memory] Saved interaction for customer '4'.

================================================================

  Customer   : Krishna (4)
  Query      : I need a refund for my annual subscription.
  Intent     : Billing
  Department : Billing
  Approval   : REJECTED
  Supervisor : Request rejected by human supervisor.
----------------------------------------------------------------
  RESPONSE:
----------------------------------------------------------------
  Hello Krishna,

  Thank you for contacting ABC Technologies Support.

  After careful review, we are unable to process your request at this time. A support specialist will follow up with you within 24
business hours to discuss your options.

  We apologize for any inconvenience and appreciate your patience.

  Best regards,
  ABC Technologies Support Team

================================================================

  Do you have another query? (yes/no) :
```

Ln 17, Col 2 Spaces: 4 UTF-8 LF { } Python venv (3.11

QUERY -5:

What was my previous support issue?

```
(venv) PS D:\Downloads\customer_support_system\customer_support_system> python demo.py
[Memory] SQLite database ready at: D:\Downloads\customer_support_system\customer_support_system\memory.db

[System] Starting AI Customer Support System...
[Graph] LangGraph workflow compiled successfully.
[System] System ready.


================================================================

  ABC TECHNOLOGIES — AI CUSTOMER SUPPORT SYSTEM
  Built with LangGraph

================================================================

  Please enter your details to begin.

  Your name       : Raju
  Your customer ID: 2

  Welcome, Raju! (ID: 2)
  Type your support query below. Type 'quit' to exit.

----------------------------------------------------------------
  Your query : What was my previous support issue?

[System] Processing your query...


[Intent] Classifying query: 'What was my previous support issue?'
[Memory] Retrieved 2 history entries for '2'.
[Intent] Classified as: Memory
[Router] Intent='Memory' → routing to 'memory_recall_agent'
[Memory Recall Agent] Retrieving conversation history...

[Supervisor] Reviewing draft response...
[Supervisor] Draft passes all quality checks ✓
[Supervisor] Notes: Draft approved without modifications.
[Memory] Saved interaction for customer '2'.
[Supervisor] Interaction saved to memory for customer '2'.

================================================================

  Customer   : Raju (2)
  Query      : What was my previous support issue?
  Intent     : Memory
```

```
  Customer   : Raju (2)
  Query      : What was my previous support issue?
  Intent     : Memory
  Department : Memory Recall
  Supervisor : Draft approved without modifications.
----------------------------------------------------------------
  RESPONSE:
----------------------------------------------------------------
  Hello Raju,

  I've retrieved your previous support history from our records.

  Your most recent support query was:
  "My application crashes whenever I upload a file."

  Full conversation history:

  === Previous Conversation History ===

  [2026-06-27T11:45:33] [Technical Support]
  Customer: My application crashes whenever I upload a file.

  [2026-06-27T11:45:33] [Technical Support]
  Agent: Hello Raju,

  Thank you for contacting ABC Technologies Technical Support.

  I understand your query: "My application crashes whenever I upload a file."

  Based on our knowledge base, here is the information relevant to your request:

  [Source: faq]
  Q: Why is my application crashing?
  A: Common causes include browser cache issues, unsupported file formats, or files exceeding size limits. Try clearing your cache
or using a different browser.

  --- Additional Troubleshooting ---
  [Source: technical_manual]
  4. FILE UPLOAD TROUBLESHOOTING
  -------------------------------
  Common causes of file upload crashes:
  - File size exceeds the plan limit (Starter: 25MB, Pro: 100MB, Enterprise: 500MB)
  - Unsupported file format (supported: PDF, DOCX, XLSX, PNG, JPG, CSV, ZIP)
  - Slow internet connection causing timeout
  - Browser cache/cookies issue

  Please do not hesitate to reach out if you need further assistance.

Best regards,
Sam | Technical Support Engineer
ABC Technologies Technical Support

=== End of History ===

Is there anything else I can help you with today?

Best regards,
ABC Technologies Support Team

================================================================

Do you have another query? (yes/no) :
```